

CV GTSRB Evaluation

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1 Introduction

The goal of this project was to test different image processing strategies and data augmentation methods on the GTSRB Traffic Sign Recognition Benchmark dataset.

GTSRB is a traffic sign classification dataset. The task is to classify each input image into the correct traffic sign class. The dataset contains 43 traffic sign classes, 39,209 training images, and 12,630 test images.

The main question of this project was:

Can image processing and augmentation strategies improve the classification performance of deep learning models on the GTSRB dataset?

To speed up the workflow, I used my own computer vision pipeline library called Vision Studio:

`https://github.com/mozi30/CV\_VisionStudio`

Vision Studio was used to make the training, inference, preprocessing, and experiment workflow easier and faster.

2 Dataset

The dataset used in this project was the German Traffic Sign Recognition Benchmark (GTSRB) [1].

It is a multi-class image classification dataset with German traffic signs. Each image belongs to one of 43 classes. The dataset is useful for testing computer vision models because traffic sign images can have different lighting conditions, image qualities, positions, and visual appearances.

During the project, it also became clear that the classes are not equally represented. Some traffic sign classes have more samples than others. Because of this class imbalance, additional training strategies such as weighted loss and balanced sampling were tested.

3 Project Tools

For this project, I used my own computer vision pipeline library:

Vision Studio
https://github.com/mozi30/CV_VisionStudio

The library was used to speed up the process of:

- loading and preparing the dataset
- applying image processing strategies
- applying augmentations
- training models
- running inference
- comparing experiments

This made it easier to test multiple strategies in a structured way.

4 Hardware

All models were trained on an RTX 5070.

The experiments were run on a remote PC. At the time of writing this documentation, I currently do not have access to the remote PC because access was lost during the weekend. Because of this, the final plots, CSV files, and full result analysis will be added later when access is available again.

5 Base Preprocessing

For the baseline and for all other tested strategies, the input images were prepared in the same basic way:

- images were resized to 224×224
- pixel values were normalized to the range 0–1

This created a consistent input format for all models.

6 Tested Image Processing Strategies

Several image processing strategies were tested to see whether they could improve classification performance.

6.1 Baseline

The baseline used only the base preprocessing:

- resize to 224×224
- normalize pixel values to 0–1

This baseline was used as the reference point for comparing all other strategies.

6.2 Histogram Equalization

Histogram equalization is a contrast enhancement technique [2] that redistributes the intensity values of an image. The method stretches the histogram of pixel intensities to make use of the full available intensity range. As a result, regions with low contrast become more distinguishable and image details can become more visible. In this experiment, histogram equalization was applied to improve the visibility of traffic sign features under varying lighting conditions.

6.3 Contrast and Brightness Normalization

Contrast and brightness normalization adjusts the intensity distribution of an image to a predefined range. The objective is to reduce variations caused by different illumination conditions, shadows, or camera exposure settings. By standardizing brightness and contrast across images, the model is presented with more consistent input data, potentially improving its ability to focus on relevant traffic sign features rather than lighting variations.

6.4 Sobel Filter

The Sobel filter is an edge detection operator [3] that calculates the gradient of image intensity in the horizontal and vertical directions. It highlights regions with strong intensity changes, which typically correspond to edges and boundaries within the image. In the context of traffic sign recognition, the Sobel filter emphasizes the outlines and structural features of traffic signs while suppressing uniform image regions.

6.5 Custom Traffic Sign Color Boost

The custom traffic sign color boost is a color enhancement technique specifically designed for traffic sign recognition. The method increases the saturation and intensity of color ranges commonly found in German traffic signs, such as red, blue, and yellow. By emphasizing these characteristic colors, the preprocessing step aims to make important sign features more prominent and easier for the model to identify.

6.6 Edge Sharpener for Large Structures

The edge sharpening method enhances prominent edges and larger structures within the image. Unlike traditional edge detection methods that only extract edge information, sharpening increases the contrast around edges while preserving the original image content. This makes the shapes and boundaries of traffic signs more distinct, potentially helping the model to better recognize sign geometry and structural characteristics.

7 Data Augmentation

In addition to image processing, data augmentation was tested during training. The following augmentations were used:

- random resized crop
- random rotation between -5° and $+5^\circ$
- random blur
- random color changes
- random brightness changes

The goal of augmentation was to create more variation during training. This can help the model generalize better because it sees slightly different versions of the same training images.

The improvement from augmentation was visible, but not very large. A likely reason is that the GTSRB dataset is already quite large and already contains different lighting conditions and visual variations. Because of this, augmentation helped, but it was not the biggest factor.

8 Class Imbalance Handling

Later in the training process, it became clear that not all classes were equally represented in the dataset.

Because of this, two methods were tested:

8.1 Weighted Loss

Weighted loss was used based on class probability. Classes with fewer samples received higher importance during training.

This helped the model pay more attention to underrepresented classes.

8.2 Balanced Sampling

Balanced sampling was also tested. The goal was to sample the training data in a way that reduces the effect of class imbalance.

Both weighted loss and balanced sampling showed an improvement compared to the baseline, especially with the custom model.

9 Models

Three model types were tested.

9.1 ResNet-50

ResNet-50 [4] was one of the primary architectures evaluated in this project. The network is based on residual learning, which enables the training of deep neural networks by mitigating the vanishing gradient problem through the use of skip connections. Due to its strong performance across a wide range of computer vision tasks, ResNet-50 was selected as a baseline architecture for traffic sign classification and served as one of the main models investigated in this work.

9.2 Swin Transformer-T

The Swin Transformer-T architecture [5] was also evaluated as part of this study. Unlike conventional convolutional neural networks, Swin Transformer employs a hierarchical transformer architecture with shifted window attention mechanisms, allowing it to efficiently capture both local and global image features. Its ability to model long-range dependencies while maintaining computational efficiency makes it a promising approach for image classification tasks such as traffic sign recognition.

9.3 Custom Hybrid Model

In addition to established architectures, a custom hybrid model was developed to explore the potential benefits of combining convolutional and transformer-based approaches within a single network. The model was designed to leverage the strengths of both paradigms:

- Convolutional layers for local feature extraction and pattern recognition.
- A lightweight transformer module for capturing global contextual information and spatial relationships within an image.

The convolutional component focuses on identifying local visual features, while the transformer component aims to improve the model’s understanding of the overall image structure. This hybrid design combines the strong local feature extraction capabilities of convolutional neural networks with the global reasoning abilities of transformer architectures.

The primary objective of this architecture was to investigate whether a lightweight hybrid approach could provide an effective balance between computational efficiency and classification performance while retaining the advantages of both CNN-based and transformer-based methods.

10 Training Pipeline

All models were initially trained using the Adam optimizer [6]. Adam was selected for the first training stage due to its adaptive learning rate mechanism and its ability to achieve fast and stable convergence during initial model optimization.

Following the base training phase, an additional fine-tuning stage was performed using Stochastic Gradient Descent (SGD) [7]. SGD is widely used for fine-tuning deep neural networks, as it can improve generalization and help models converge to more robust solutions.

The training pipeline therefore consisted of two optimization stages:

- Initial training with the Adam optimizer.
- Fine-tuning with the SGD optimizer.

The purpose of this approach was to combine the rapid convergence properties of Adam with the strong generalization capabilities commonly associated with SGD. Furthermore, it enabled a direct comparison of the effectiveness of both optimizers during the later stages of model training.

For most experiments, the fine-tuning stage was performed using SGD. The Swin Transformer-T architecture represented a special case, as its performance stabilized during the initial training phase, reducing the need for extensive additional optimization.

11 Results

11.1 General Model Performance

11.2 Image Processing Results

The image processing strategies often made the performance worse instead of better.

The likely reason is that some processing methods removed useful information from the image or created unnatural inputs. Modern deep learning models can often learn useful features directly from the original image, so strong image processing can sometimes hurt instead of help.

Histogram equalization had an improvement in all models. A likely cause is that it also functions as a kind of normalisation of the the input.

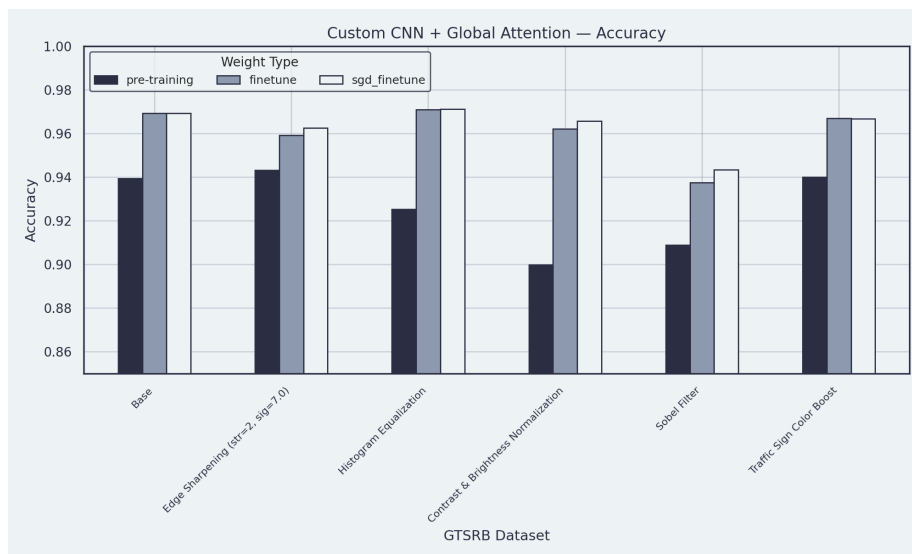


Figure 1: Custom Model Accuracy

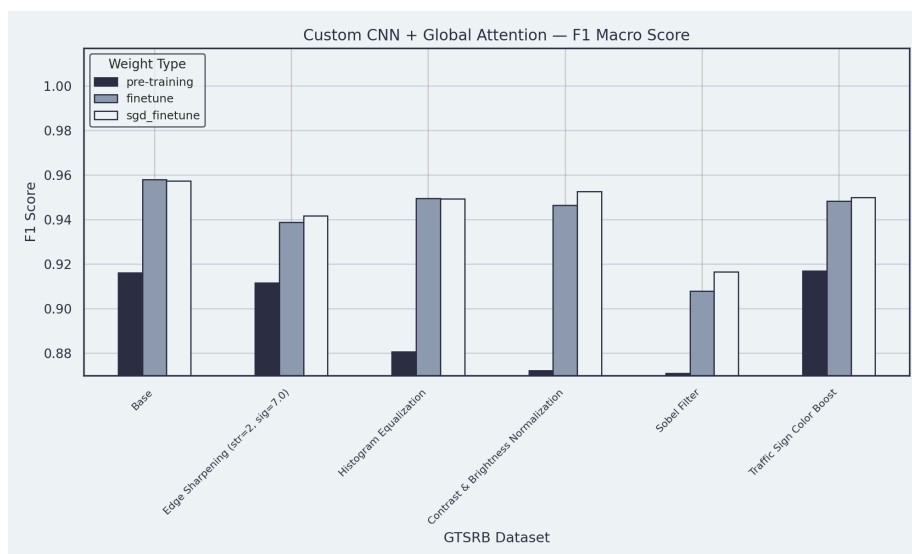


Figure 2: Custom Model F1 Macro Score

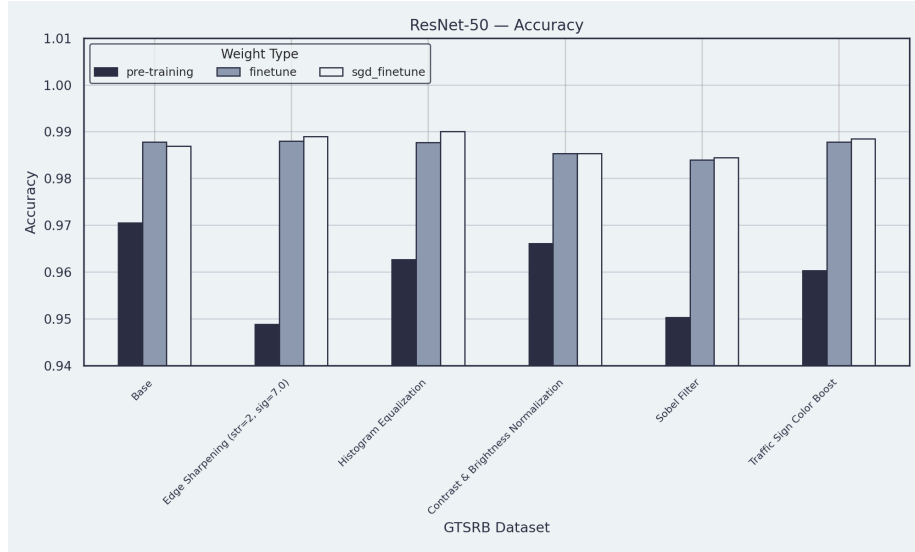


Figure 3: ResNet50 Accuracy

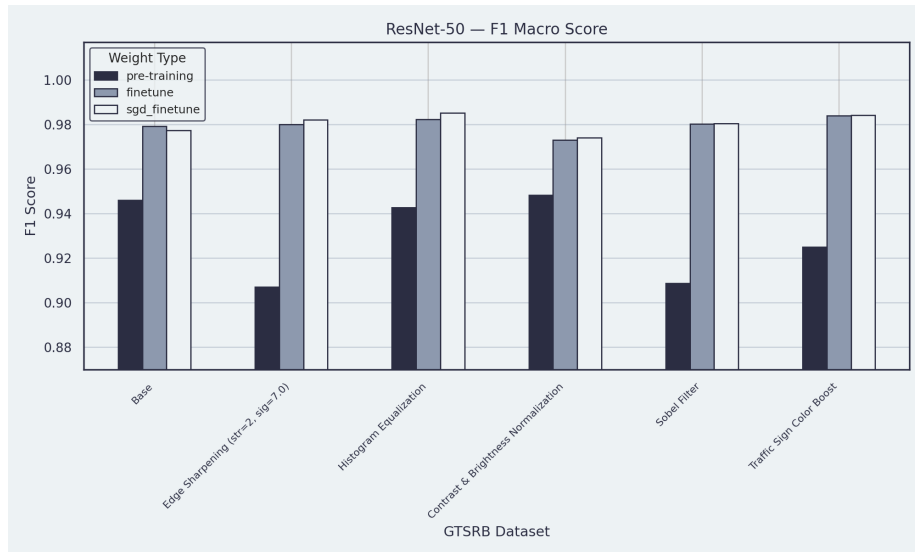


Figure 4: ResNet50 F1 Macro Score

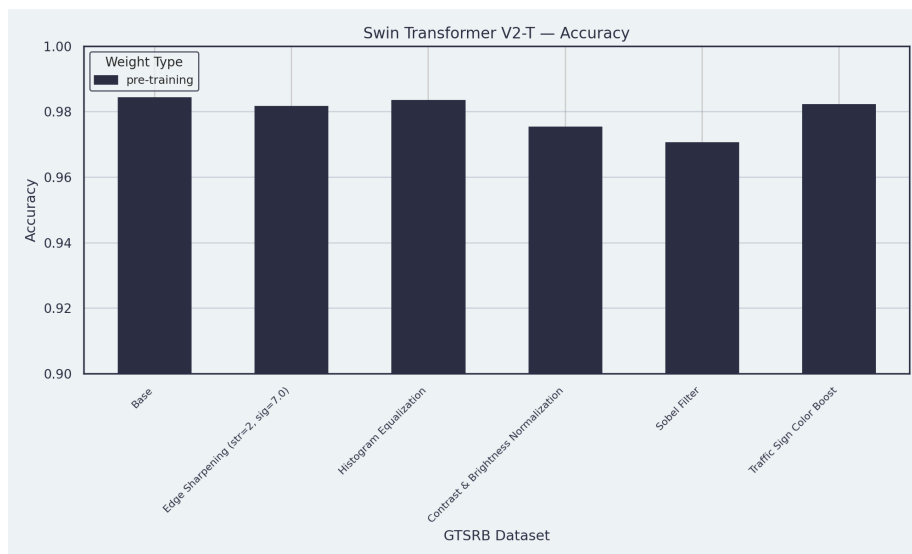


Figure 5: SwinTransformer Accuracy

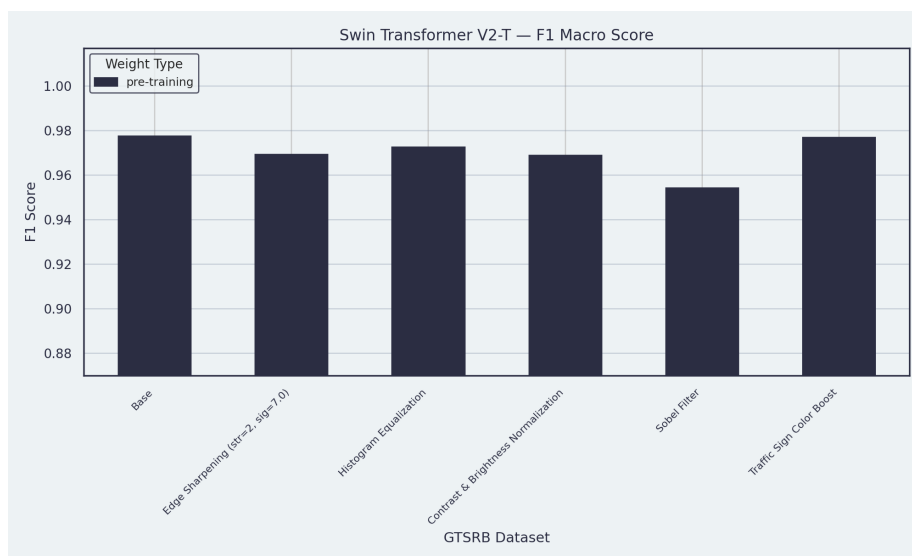


Figure 6: SwinTransformer F1 Macro Score

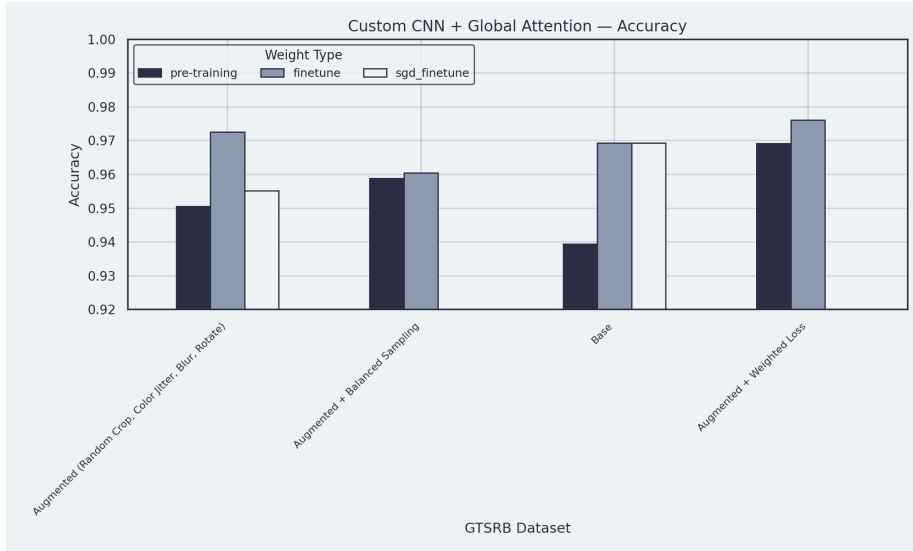


Figure 7: Custom Model Augmentation and Class Imbalance Accuracy

11.3 Augmentation Results

Training with augmentation showed a small improvement compared to the baseline. As it can be seen in 7 and 8 the from augmentation improvement was not huge, probably because GTSRB is already a large dataset with many different lighting conditions and image variations.

Still, light augmentation helped create more varied training samples and improved robustness.

11.4 Class Imbalance Results

After discovering the class imbalance, weighted loss and balanced sampling were tested.

While the weighted loss improved the models performance the balanced sampling did not, likely since the validation set has the same class distribution as the training set nudging performance to overrepresented classes.

This showed that checking class balance is important before and during training.

12 Final Best Model Experiments

The information from all previous experiments was used to create the best possible model setup. Since ResNet-50 was the best general model, it was selected for the final experiments.

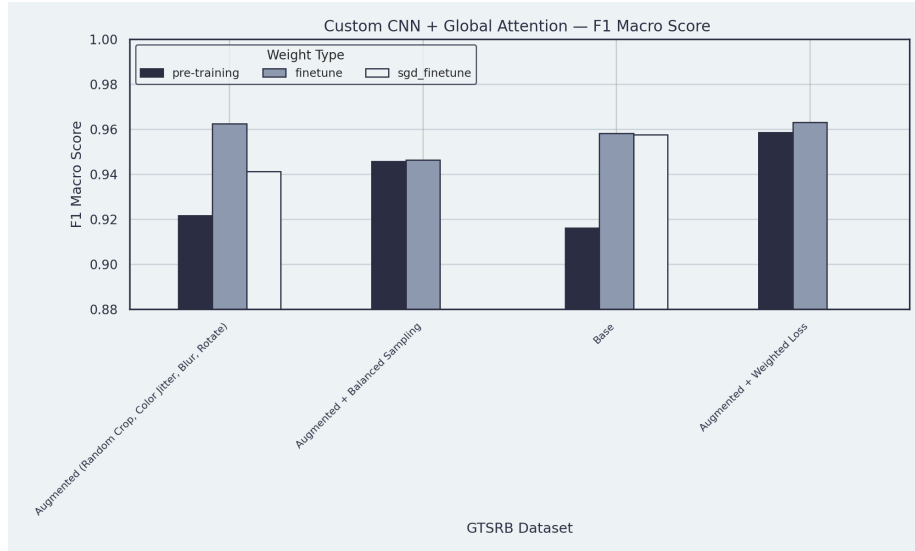


Figure 8: Custom Model Augmentation and Class Imbalance F1 Macro Score

Two final ResNet-50 setups were trained.

12.1 ResNet-50 with Weighted Loss

This setup used:

- ResNet-50
- histogram equalization
- augmentations
- weighted loss
- base training
- SGD fine-tuning

12.2 ResNet-50 with Balanced Sampling

This setup used:

- ResNet-50
- histogram equalization
- augmentations

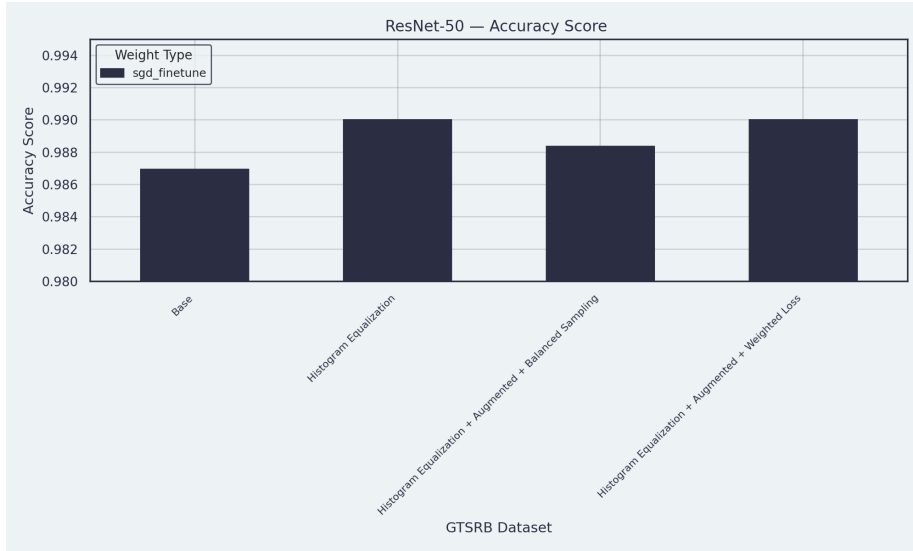


Figure 9: ResNet-50 best model Accuracy

- balanced sampling
- base training
- SGD fine-tuning

As seen in Figure 9 and 10, adding all performance increasing optimizations together did not bring the hoped-for combined performance increase. Adding augmentation and weighted loss to histogram equalization did not provide any performance boost on the validation set, while balanced sampling even made the performance both on accuracy and F1 score worse. Overall reaching an accuracy of 99% is still a very good score for this dataset.

13 Model Ensembles

In the second part of the experiment, model ensembles are evaluated. Model ensembles combine the predictions of multiple models to produce a final decision [8]. The underlying assumption is that different models focus on different features of the input image and therefore exhibit distinct strengths and weaknesses. By combining their predictions, individual errors can be reduced, resulting in a more robust and reliable classifier.

To investigate the effectiveness of ensemble methods, three different approaches for combining model outputs are explored:

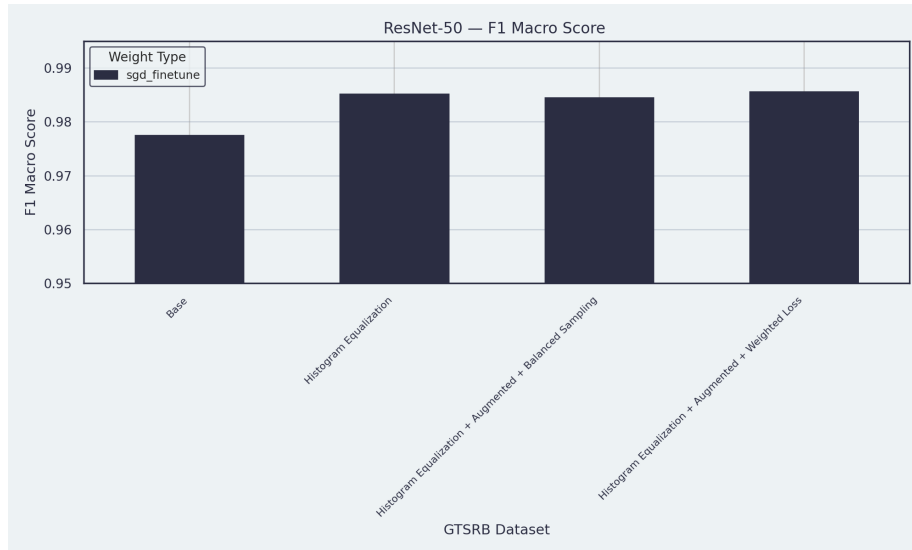


Figure 10: ResNet-50 best model F1 Macro Score

13.1 Soft Voting

Soft voting combines the predicted class probabilities of multiple models by averaging them. The final prediction is then determined by selecting the class with the highest average probability. Unlike hard voting, soft voting takes the confidence of each model into account, allowing highly confident predictions to have a stronger influence on the final decision. This often results in better performance, especially when the individual models are well-calibrated.

13.2 Weighted Soft Voting

Weighted soft voting extends the soft voting approach by assigning a weight to each model based on its individual performance. Models with higher validation accuracy contribute more strongly to the final prediction, while weaker models have less influence. The final class probabilities are calculated as a weighted average of the individual model probabilities. This approach can further improve ensemble performance by leveraging the strengths of the best-performing models.

13.3 Hard Voting

Hard voting combines the predictions of multiple models by considering only their predicted class labels. Each model casts one vote for its predicted class, and the class receiving the majority of votes is selected as the final prediction. While

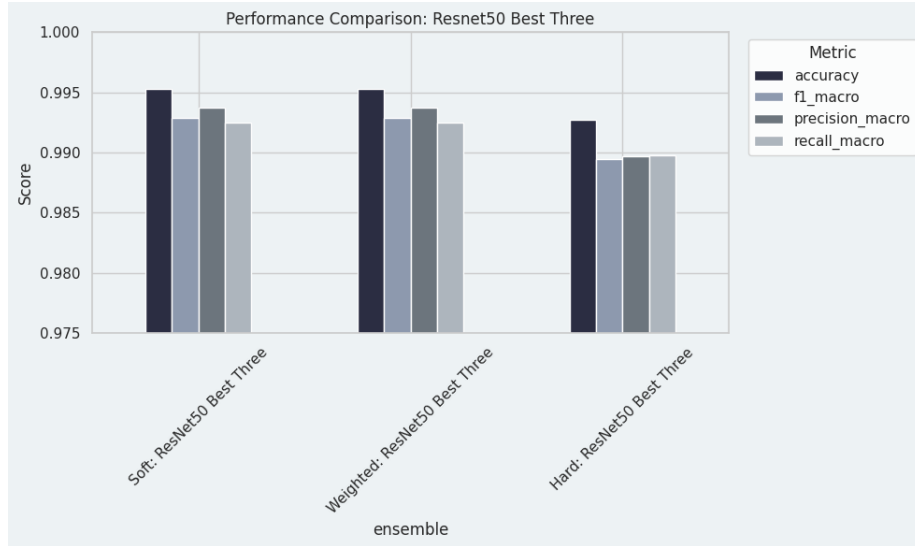


Figure 11: ResNet-50 ensemble results

this method is simple and computationally efficient, it ignores the confidence of individual models and may therefore perform worse than soft voting approaches.

13.4 Results

First in figure 11 12 13 the best 3 performing configurations of each model are combined to one classifier. This is done to see how combining the same base architecture combines to form a stronger detector.

To gain further insight, the best performing model of each model architecture is combined as seen in figure 14. Since every model processes and focuses on different features in the image, the expectation is that combining them will filter each models weaknesses and boost overall robustness. For further insight all the models are used combined in a single ensemble for classification. Therefore 9 different models vote for the final result, to arrive to a common result. The result can be seen in figure 15.

14 Discussion

The experiments showed that image processing does not automatically improve model performance.

Some processing strategies can make images look cleaner or more structured for humans, but they may remove important information for the model. This was especially visible with Sobel filtering, which likely removed too much fine

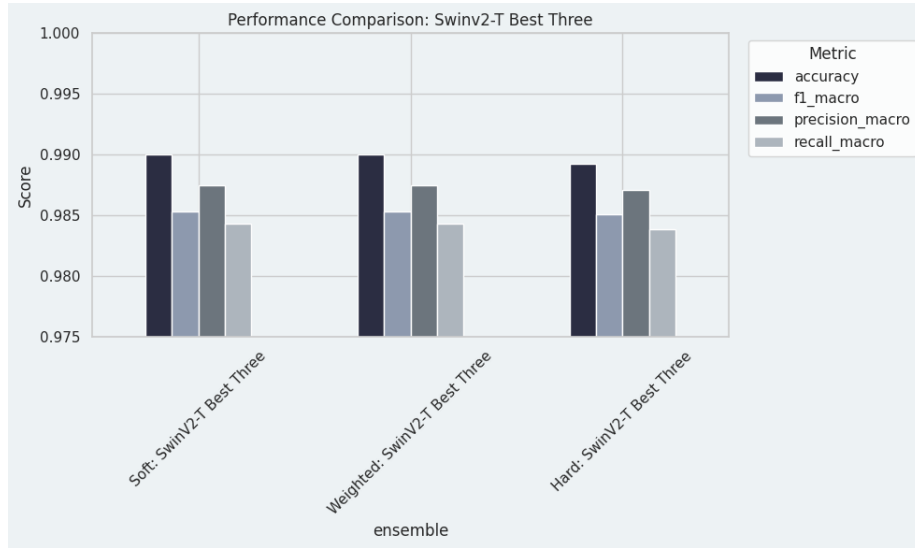


Figure 12: SwinTransformer ensemble results

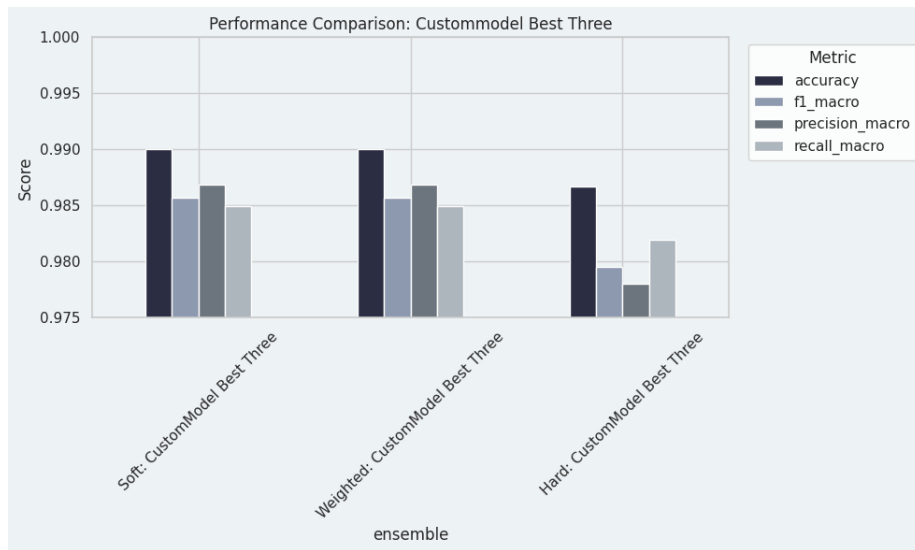


Figure 13: Custom model ensemble results

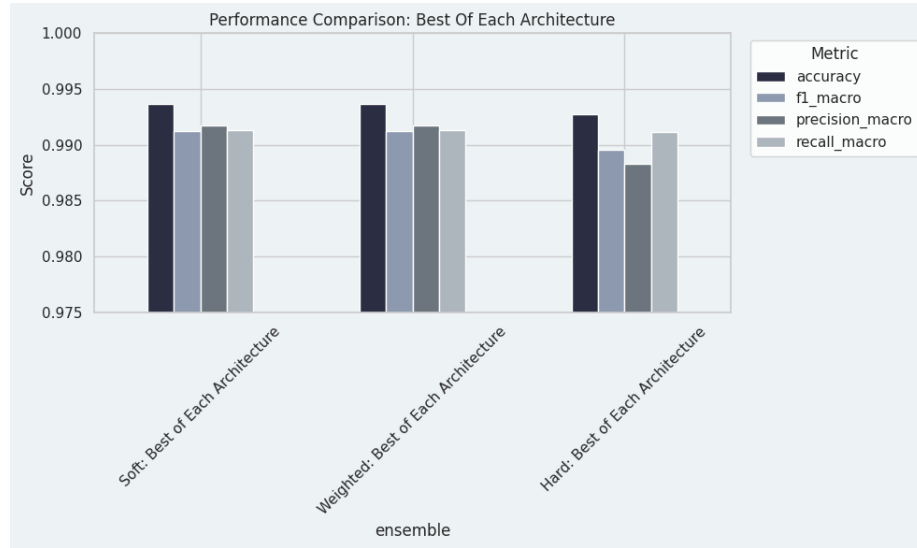


Figure 14: Best model from each architecture ensemble results

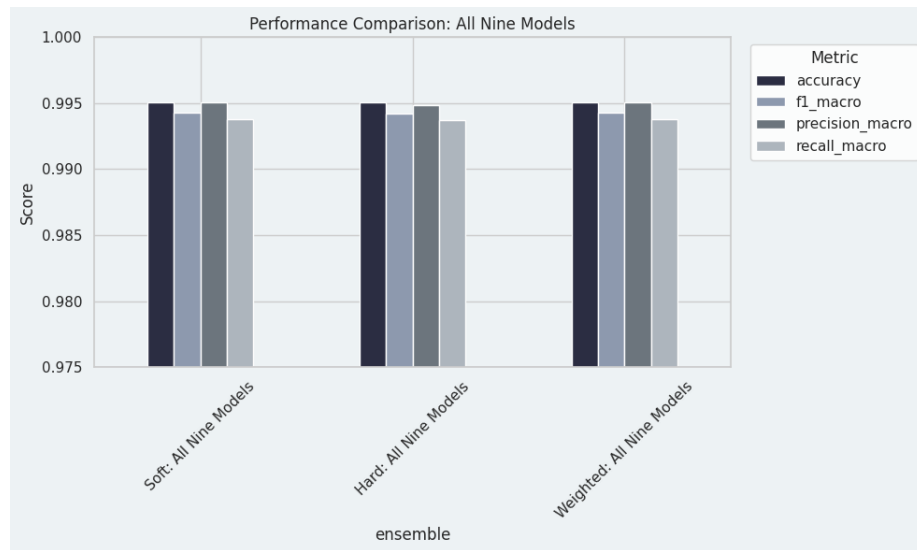


Figure 15: All models ensemble results

information from the image.

The best image processing method was histogram equalization. This probably helped because it made contrast and lighting more consistent across images. However, it also adds extra preprocessing time during inference. In an online or real-time system, this additional inference cost could become a disadvantage.

Augmentation gave a small improvement. This makes sense because augmentation creates more variation during training. However, because the GT-SRB dataset is already quite large and contains different lighting conditions, the improvement was not as strong as it might be on a smaller dataset.

The class imbalance experiments were very useful. Weighted loss and balanced sampling both improved performance, showing that class distribution should always be checked before training.

The custom model was also an important part of the project. Even though it did not outperform ResNet-50 or Swin Transformer-T, it achieved competitive performance while being smaller and faster. This shows that custom models can be useful when designed around the requirements of the task.

The results demonstrate that ensemble methods improve detection performance across all tested configurations. Although models based on the same architecture exhibit comparable individual performance, as shown in figures 11, 12, and 13, combining model outputs consistently improves overall accuracy. These findings support the hypothesis that ensemble learning enhances both robustness and predictive performance by mitigating the weaknesses of individual models.

Combining the best-performing model from each architecture also yields better results than using any single model independently. However, the benefits of architectural diversity and different feature extraction strategies do not outweigh the lower detection performance of the less accurate models. Consequently, an ensemble consisting of three ResNet50 models achieves better performance than the heterogeneous ensemble of the best model from each architecture. This effect is clearly visible in 14.

The strongest performance is achieved by combining all nine trained models into a single ensemble. While this approach requires the greatest computational resources, it produces the best overall results. As seen in 15, both accuracy and F1-score approach 99.5% across all three voting strategies. This performance comes close to the best reported competition accuracy of 99.82%, highlighting the effectiveness of large-scale ensemble methods for this classification task.

15 What I Learned

This project gave several important insights.

First, image processing with sophisticated deep learning models can make performance worse if it creates unnatural images or removes important information. Most of the time, preprocessing should be light and should not destroy useful image details.

Second, histogram equalization was useful because it acted like a form of normalization for image contrast. However, it also increases inference time because every image has to be preprocessed.

Third, class balance is very important. The class imbalance in the dataset affected training, and methods like weighted loss and balanced sampling helped improve performance.

Fourth, light augmentation can improve performance by creating more varied training samples. This is especially useful for underrepresented classes.

Finally, creating a custom model can still be valuable. My custom hybrid model was smaller and faster than ResNet-50 and Swin Transformer-T, while still reaching competitive accuracy. This showed that a model designed for the specific problem can be useful even if it does not reach the absolute highest score.

Model ensembles often achieve higher performance than individual models by leveraging the strengths of multiple predictors. Combining the class probability distributions of several models increases robustness and helps mitigate the weaknesses of any single model. This approach can lead to more reliable and accurate predictions, particularly in domains such as medicine, where maximizing predictive performance is often more important than real-time inference speed. As a result, ensemble methods can have a significant positive impact on diagnostic and decision-support outcomes.

16 Conclusion

This project tested different image processing strategies, augmentations, model architectures, and class imbalance methods on the GTSRB Traffic Sign dataset.

The experiments showed that strong image processing does not always help. In many cases, it made performance worse because useful image information was removed. Histogram equalization was the only image processing method that improved performance clearly.

Augmentation gave a small improvement, while class imbalance handling with weighted loss and balanced sampling gave useful performance gains.

The best final approach was based on ResNet-50 using histogram equalization, augmentation, class imbalance handling, and SGD fine-tuning. This setup reached slightly above 99% accuracy and 98.5% F1-score.

For a balanced computation performance to classification performance the best outcome is to use soft voting of the 3 best ResNet50 models providing a accuracy of 99.5%.

Overall, the project gave useful insight into how preprocessing, augmentation, optimizer choice, class imbalance handling and model ensembles can affect traffic sign classification performance.

17 References

References

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- GTSRB dataset information: <https://paperswithcode.com/dataset/gtsrb>
 - Vision Studio repository: https://github.com/mozi30/CV_VisionStudio